

EW – 1 Lab 4

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VOLTAGE REGULATOR (LTspice):

A Zener diode is a specially built diode; its depletion layer breaks down (in reverse bias) at low voltages and the characteristic curve post the breakdown voltage shows minimal voltage changes for large current changes. Thus, it acts as a constant voltage regulator over a good range. However, it sometimes generates electrical noise on top of the DC supply. A decoupling capacitor of large value is connected across the output to filter the noise.

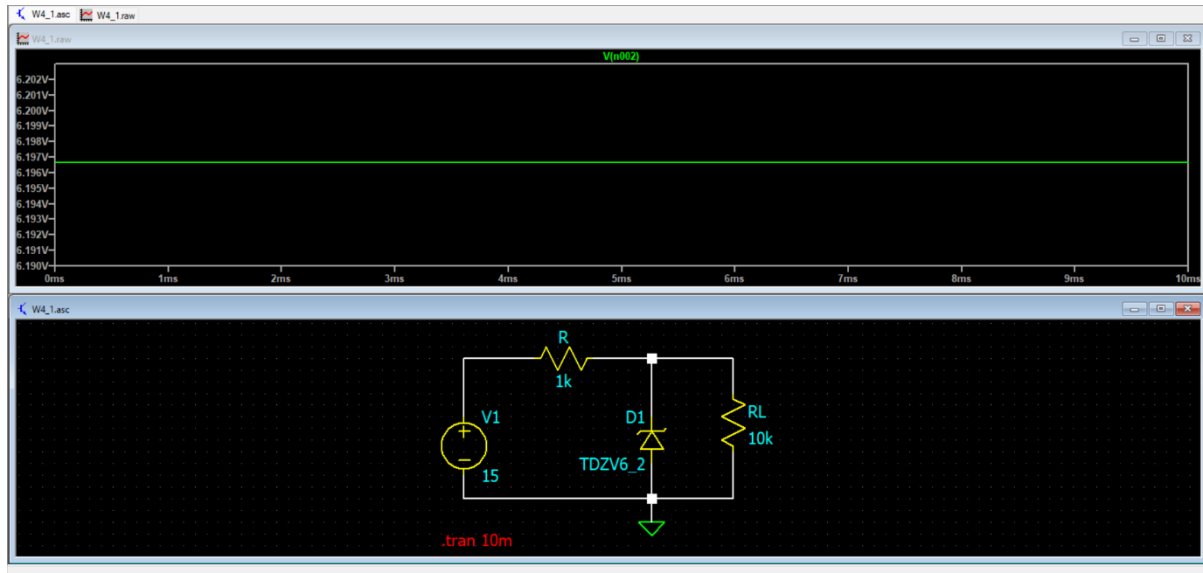
The LM7805 IC is a regulator package that outputs a constant 5V potential over a good range of input voltages. Just like a Zener diode, it too may generate noise and so, a decoupling capacitor is used in a similar fashion.

AIM:

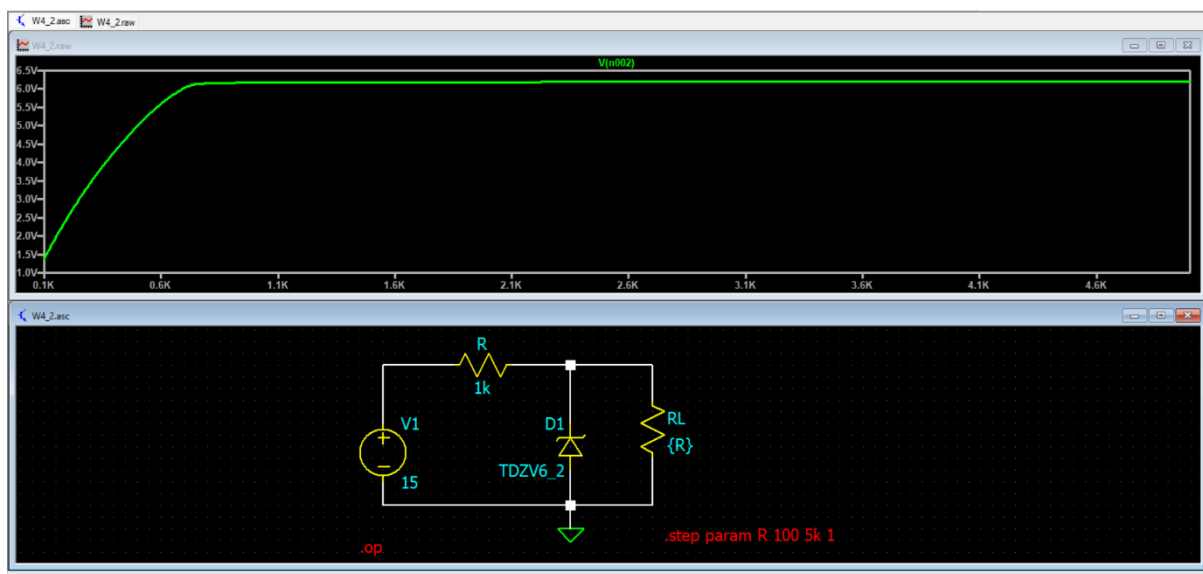
- Design a simple voltage regulator circuit using a Zener diode.
- Contrast the performance of this circuit with an IC voltage regulator.
- Check the noise performance of this circuit.

Results:

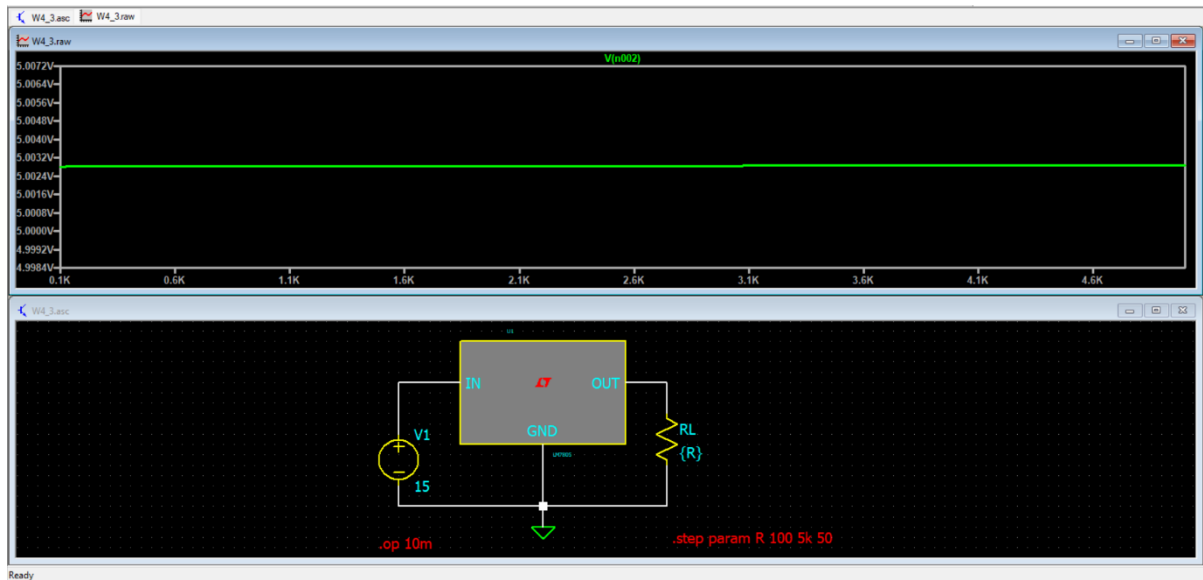
Zener Diode voltage regulator (transient analysis)



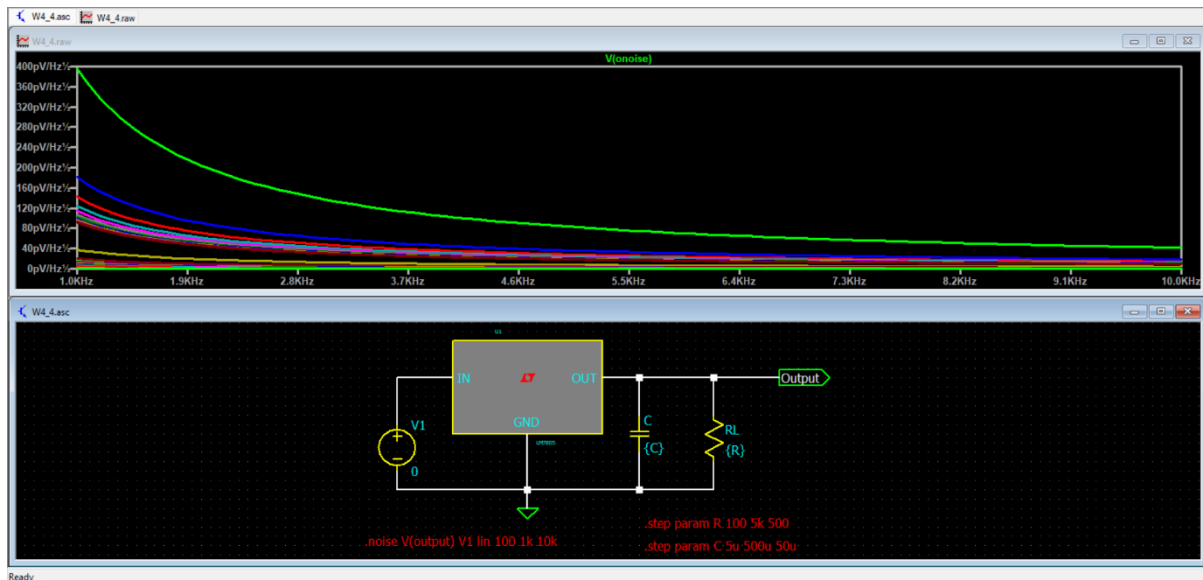
Zener Diode voltage regulator (V_{out} v/s R_L)

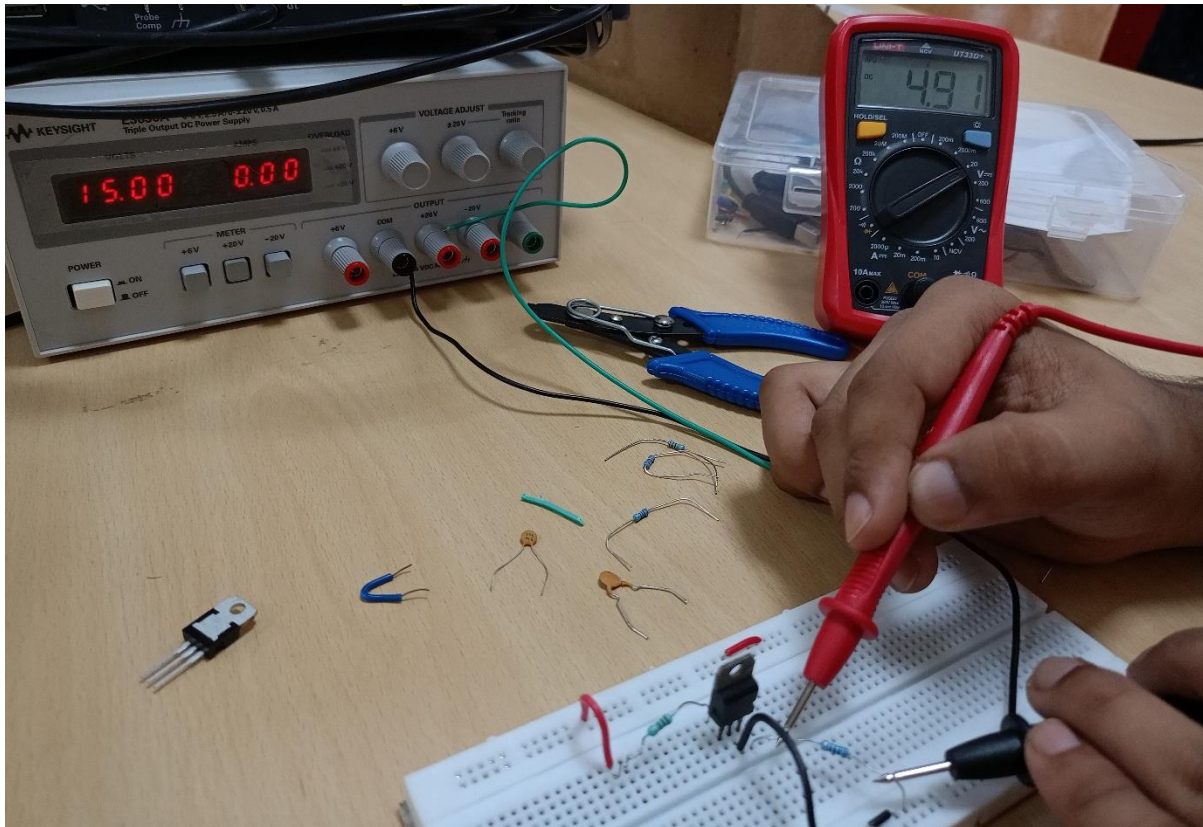


LM7805 Voltage Regulator IC (load regulation)



LM7805 with decoupling capacitor (noise analysis)





Observations:

- At $R_L = 10\text{ k}\Omega$, the Zener Diode regulator circuit gave a constant output of $\sim 6.19\text{ V}$. That is, at sufficiently large loads, the diode maintains a constant (breakdown) voltage.
- The V_{out} v/s R_L graph shows that around a resistance of $650\text{--}750\ \Omega$, the output voltage becomes constant, before which it varied with load resistance. This is because for smaller loads the Zener doesn't breakdown (smaller loads act as shunt and allow large currents through). Investigation shows that at $R_L = 704.5\ \Omega$, an ideal Zener diode will break down; the breakdown voltage being 6.2 V .
- With the LM7805 IC replacing the Zener diode, load regulation is seen, i.e., the (simulated) output voltage is constant at 5 V for all values of R_L . In comparison, the actual output came out to be 4.91 V .
- A noise analysis shows how higher noise frequencies bypass the capacitor, while lower frequencies cannot penetrate the capacitor. This is due to high resistance offered by the capacitor to low frequency noise. As the capacitance is increased, greater noise cancellation is achieved.

Conclusion: The voltage regulator circuits were successfully analysed.